CLASS X - COORDINATE GEOMETRY

Ultimate Practice Session

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Quick Fire

The distance of the point P(4, −3) from the origin is:

(a) 1 unit

(b) 7 units

(c) 5 units

(d) 3 units

Ans: (c) 5 units

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Quick Fire

The distance between the points A(2, −3) and B(2, 2) is:

(a) 2 units

(b) 4 units

(c) 5 units

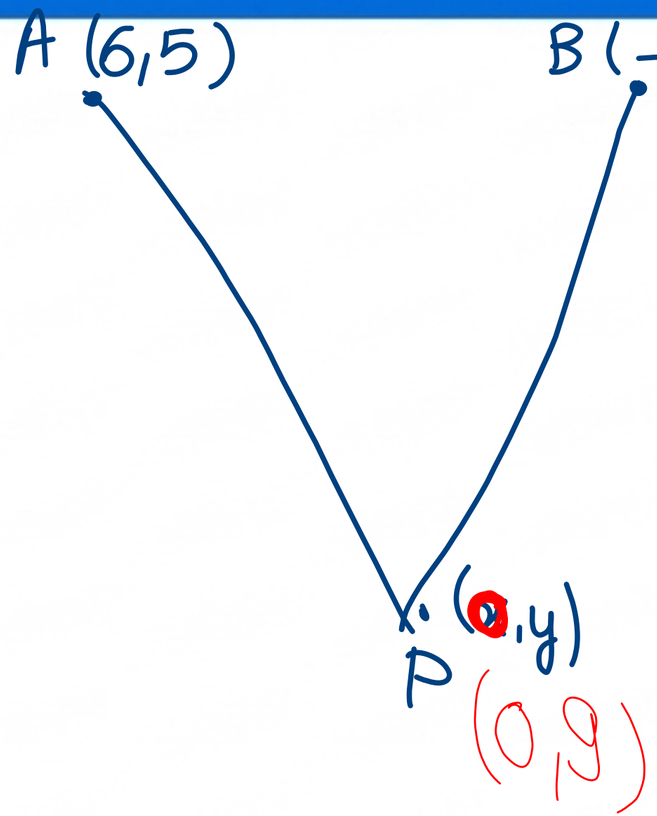
(d) 3 units

Ans: (c) 5 units

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Practice

Find a point on the y-axis which is equidistant from the points A(6, 5) and B(– 4, 3).



$AP = BP$

$$\sqrt{36+(y-5)^2} = \sqrt{16+(y-3)^2}$$

Squaring both sides:

$$\Rightarrow 36+y^2+25-10y = 16+y^2+9-6y$$

$$61-10y = 25-6y$$

$$36 = 4y$$

$$y = 9$$

Ans: (0, 9)

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Brain Teaser

Find all the possible value of a for which the distance between the points A(a, –1) and B(5, 3) is 5 units.

$AB = 5$ units

$$\sqrt{(5-a)^2+16} = 5$$

$$\sqrt{25+a^2-10a+16} = 5$$

$$1a^2 - 10a + 16 = 0$$

$ax^2+bx+c=0$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$a = \frac{-(-10) \pm \sqrt{100-4(1)(16)}}{2}$$

$$= \frac{10 \pm \sqrt{36}}{2} = \frac{10 \pm 6}{2}$$

$$\frac{10+6}{2} = 8$$

$$\frac{10-6}{2} = 2$$

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💪 Practice

Find the values of y for which the distance between the points P(2, – 3) and Q(10, y) is 10 units.

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Quick Fire

The point P(1, 2) divides the join of A(–2, 1) and B(7, 4) in the ratio of:

(a) 3 : 2 (b) 2 : 3 (c) 2 : 1 (d) 1 : 2

$$x = \frac{m(7) + n(-2)}{m+n}$$

$$7m - 2n = m + n$$

$$6m = 3n$$

$$\frac{m}{n} = \frac{3}{6}$$

$$m : n = 1 : 2$$

Ans: (d) 1 : 2

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Quick Fire

A point P divides the join of A(5, –2) and B(9, 6) in the ratio 3 : 1. The coordinates of P are:

(a) (4, 7)

(b) (8, 4)

(c) (11/2, 5)

(d) (12, 8)

Ans: (b) (8, 4)

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💪 Practice

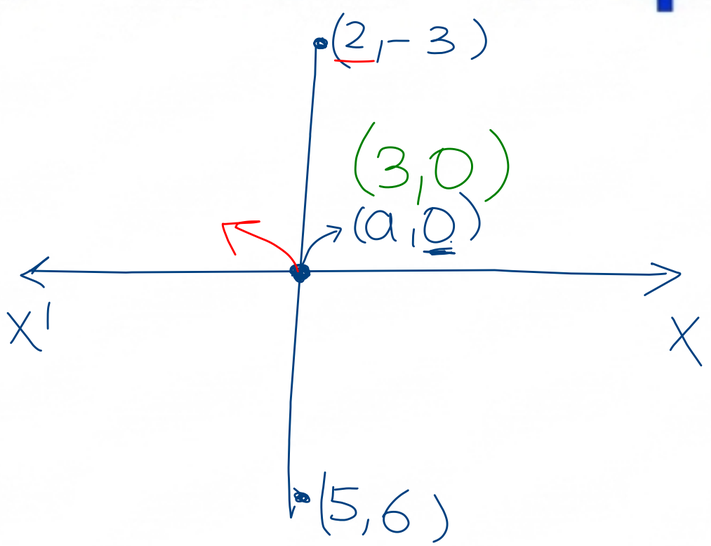
Find the coordinates of the point which divides the line segment joining the points A(4, −3) and B(9, 7) in the ratio 3 : 2.

Ans: (7, 3)

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High Yield

In what ratio is the line segment joining A(2, -3) and B(5, 6) divided by the x-axis? Also find the coordinates of the point of intersection.



$$x = \frac{mx_2 + nx_1}{m+n} \quad \& \quad y = \frac{my_2 + ny_1}{m+n}$$

$$a = \frac{1(5)+2(2)}{1+2}$$

$$a = \frac{9}{3}$$

$$\boxed{a = 3}$$

$$0 = \frac{m(6)+n(-3)}{m+n}$$

$$6m - 3n = 0$$

$$6m = 3n$$

$$\frac{m}{n} = \frac{3}{6}$$

$$\boxed{m:n = 1:2}$$

Ans: Ratio 1:2, Point (3, 0)

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High Yield

If the point C(−1, 2) divides the line segment AB in the ratio 3 : 4, where the coordinates of A are (2, 5), find the coordinates of B.

Ans: B(-5, -2)

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🏆 Final Boss

If P(1, 2), Q(4, 6), R(5, 7) and S(a, b) are the vertices of a parallelogram PQRS, find the values of a and b.

Ans: a = 2, b = 3

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